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Module 1: Search Skills & Open Source Intelligence (OSINT)

1. File & Threat Intelligence Analysis using VirusTotal

Concept: VirusTotal acts as an aggregation platform that consolidates detection data from over 70 antivirus vendors and web scanning tools. It allows security analysts to quickly inspect suspicious files, hashes, domain names, and URLs without running the files directly.

Practical Application:

Submitted a sample executable file (invoice_payment.exe) to the simulated VirusTotal scanner.

Reviewed the detection engine metrics at the top of the interface to evaluate the consensus among global security vendors regarding the file's malicious nature.

Key Takeaway: Using threat intelligence tools helps blue team analysts accelerate incident response by verifying whether a file is a known threat.

2. Understanding Vulnerability Databases (CVE & CVSS)

Concept: The Common Vulnerabilities and Exposures (CVE) system provides a standardized naming convention for publicly disclosed software security bugs. Each entry follows the format CVE-YEAR-NUMBER (e.g., CVE-2026-1337).

Scoring Framework: Vulnerabilities are rated using the Common Vulnerability Scoring System (CVSS) based on three primary factors:

Impact: The degree of potential damage.
Complexity: How easy or difficult it is to execute the exploit.
Availability: The likelihood of exploitability across public environments.

Practical Application: Queried the vulnerability database for CVE-2026-1337 to analyze its severity rating and examine Proof of Concepts (PoCs) on repositories like ExploitDB.

Key Takeaway: Vulnerability management relies on CVSS scoring to prioritize patching critical vulnerabilities before lower-risk ones.

3. Command Line Documentation (Linux MAN Pages)

Concept: Official product documentation and built-in system manuals are the most accurate reference materials for understanding command-line tools.

Practical Application:

Accessed built-in Linux manual pages using the command format man <command> (e.g., man nc for Netcat).

Analyzed command syntax, parameter flags, and practical usage examples, such as establishing network connections to host destinations on specific target ports (e.g., nc host.example.com 42).

Key Takeaway: Mastering man pages reduces reliance on third-party guides and ensures correct syntax execution in CLI environments.

4. Security Research & Code Verification via GitHub

Concept: GitHub is widely used by security researchers to publish vulnerability reports, scanner scripts, and Proof of Concept (PoC) exploit code.

Practical Application:

Searched GitHub for target CVE identifiers (e.g., CVE-2026-1337).

Reviewed the README.md documentation to identify the primary demonstration script before running any code.

Key Takeaway: Always audit third-party code on GitHub prior to execution, as unverified repositories may contain broken or malicious codes

Module 2: Learning Cyber Security Essentials

1. Web Application Security

Concept: Securing web applications requires understanding client-server communication, authentication mechanisms, and common web flaws.

Practical Application:

- Interacted with a simulated vulnerable web application (BookFace).
- Identified the target account username (Ben.Spring).

- Utilized an automated authentication test (brute-forcing 4-digit PINs) to recover access and achieve account takeover.

- Key Takeaway: Rate limiting and strong authentication controls are essential to defend against automated account takeover attacks.

2. Network Security

Concept: Network security involves inspecting packet traffic, monitoring log events, and ensuring system availability across organizational boundaries.

Practical Application:

1. Reviewed historical security breach case studies (e.g., the Target corporate data breach).

2. Analyzed how lateral movement and unmonitored network vectors lead to massive data exfiltration costing hundreds of millions of dollars.

3. Key Takeaway: Continuous log monitoring and network segmentation are critical for reducing total breach impact.

3. Cyber Security Specialization Pathways

Offensive Pentesting (Red Teaming): Focuses on ethical hacking, system exploitation, vulnerability assessment, and offensive security testing.

Cyber Defense (Blue Teaming): Focuses on Security Operations Center (SOC) monitoring, threat hunting, forensic investigation, and incident response.